

Velocity measurement of a ball in motion via intensity-based kinematics

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Abstract

Video processing is relatively similar to image processing but with the involvement of frame rate. Some limitations can be observed after applying morphological operations and image segmentation during the image processing part of video processing. A different method is proposed by using the intensity of a ball to measure its position relative to the camera. The velocity of the moving ball can be achieved through video processing and calculus. The ball will be rolled twice; one trial will be rolled relatively slower compared to the other to create contrast and basis for results. A position vs. time graph will be plotted for both cases. The derivative of the function from the graph plotted will be the velocity of the ball at that particular case. The goal of the study is to solve real world scenarios using calculus lens. The velocity of the ball measured by intensity-based kinematics and yielded velocities equal to 0.0164 m/s for a slow ball and 0.0349 m/s for a fast ball each having percent errors equal to 4.65% and 33.21%, respectively.

1. Introduction

It is widely known that a video is a collection of images (or frames) captured by an imaging device at a certain frame rate. The frame rate defines how fast the imaging device captures frames per second [1]. From the frame rate, the time interval in between images (Δt) can be extracted and used in video processing. Equation 1 shows the equation to solve for the said time interval.

$$\Delta t = \frac{1}{\text{framerate}} \quad (1)$$

Video processing is the process of analyzing a video by separating it into different frames. A more popular application of video processing is solving kinematics-based problems. When the video file is converted to image files, image processing can be performed to extract the desired data. Kinematics experiments (measurement of gravity,

pendulums, etc.) can now be performed as the time factor can now be estimated using the frame rate of the imaging device.

Since video processing is similar to image processing with the involvement of the frame rate, the process still comes down to the limitations of image processing. Some errors may occur while performing morphological operations [2] or image segmentation [3] like that shown in Figures 1 and 2, respectively.



Figure 1: An error that would occur in performing morphological operations.



Figure 2: An error that would occur in performing image segmentation.

The errors observed above can be removed by being more specific in the morphological operations and image segmentation but this would become a very big problem in the automation process of the program. The change would also affect the other images in the frames of the video and it would most likely increase the run time of the program.

The shapes in Figure 1 and 2 would have a great effect in deviation of the mean of the supposed-to-be circles. This would then lead to a higher margin of error. A solution to the stated problem is the implementation of intensity-based kinematic. Intensity does not greatly vary when some parts of the ball is removed as long as a reasonable amount of the ball is still present. This will give a big advantage over the normal image processing method in terms of margin of error.

The intensity of an object (I_o) as measured by the imaging device is inversely proportional to the square of its distance from the object (r) [4] as shown in equation 2. An inverse square law is a characteristic of anything that spreads out in straight lines, without getting lost. Both gravity and electric also share the same property. A proportionality constant (k) can be added to the relation and by a few manipulations, equation 3 can be used to show a linear relation to intensity and distance.

$$I_o \propto \frac{1}{r^2} \quad (2)$$

$$\sqrt{I_o^{-1}} = kr \quad (3)$$

To obtain the intensity of an object digitally, equation 4 will be used where R, G and B are the values for the respective red, green and blue channels (0-255). The resulting value for I_o will be from 0-1.

$$I_o = \frac{R+G+B}{765} \quad (4)$$

Based on the intensity of an object at a certain distance, the position (distance from the camera) can be estimated. After getting the position, the estimated velocity can then be solved via video processing and using derivative. The objective of this study is to measure the velocity of a moving ball via intensity-based kinematics and calculus.

What are derivatives?

In mathematics, the derivative of a function measures the sensitivity to change of the function value with respect to a change in its argument. A derivative is a way to show instantaneous rate of change: that is, the amount by which a function is changing at one given point. In our case, we're only going to be dealing with simple linear equations. This will make our lives easier since the derivatives of linear functions are constant.

To find the derivative of a function, we will need to use the slope limit formula. Then we make Δx approach to 0. This makes the distance between the two points selected shrink to zero. The smaller the difference between the two points, the closer we get to the value of the slope at point 1. Therefore, if Δx is infinitesimally small (i.e. 0.000...001), the value of the slope will be very precise.

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Personal Connection

I elected to conduct this study because I've always had a passion towards things that are in enigma. Physics played a major influence towards tinkering around this research topic. Quantum mechanics, in specific, tickled my curiosity towards studying the

behaviors of light. As an aside to this research topic, using the intensity of light can also have potential in using as a means to detecting the distance of an object. This study is significant towards the real world as well. Self-driving cars use image segmentation to detect entities surrounding it. This makes learning how to use it intriguing and engaging to use in the investigation.

2. Methodology

Experimental Setup

The setup for the experiment is shown in Figure 3. The experiment was conducted in a dark room such that no stray light will contribute to the intensity of the object. The light source is a white LED flashlight shone from behind and above the camera. The object used is a ball with a solid color to remove variation in intensities and the imaging device used to capture the experiment is a GoPro HERO5 Black camera with a frame rate of 30 frames/second and set the Auto Low Light to *OFF*.

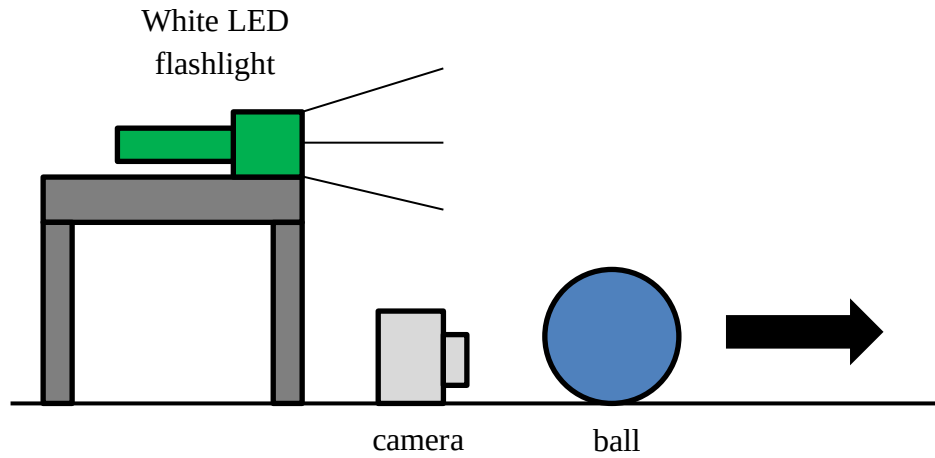


Figure 3: The setup used for the experiment.

Calibration

Before performing the experiment, a calibration of the intensity to position relation is performed. The ball was placed at a known distance from the camera while capturing its intensity. The intensity of the ball was measured at positions equal to 0.35m to 1.50m from the camera at increments of 0.05m. A curve was plotted from the relation $\sqrt{I_o^{-1}}$ to the position (r) of the ball. Figure 4 shows the said plot.

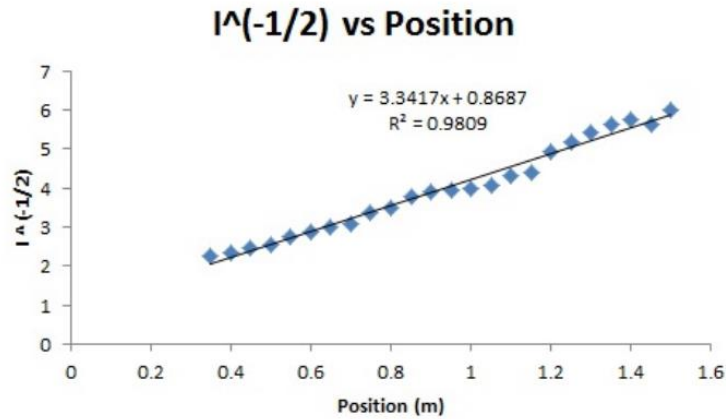


Figure 4: Calibration curve for the relative intensity vs. position.

With an R^2 value of 0.98, the graph can be used as a good estimate to the relation of intensity to position. Equation 5 shows the equation for the relation.

$$\sqrt{I_o^{-1}} = 3.3417 + 0.8687 \quad (5)$$

Experiment

For the experiment proper, the ball was allowed to roll from the camera going further away from it. The ball was pushed by a human and made sure it followed a straight path to remove further sources of errors. Two experiments were performed: one with the ball moving fast and the other with the ball moving slower relative to the other trial. The time it takes for the ball to reach a certain point is measured using a stopwatch recorded by an observer. A position vs. time graph will be plotted for both cases and a line will be fitted to the curve. The slopes of both lines will be the velocity of the ball at that particular case.

3. Results and Discussion

It is known that the frame rate of the camera is 30 frames/second. Using equation 1, the time interval in between consecutive frames is calculated and is equal to $0.03\overline{3}$ seconds.

For the case of the fast moving ball, only 84 frames captures the ball in motion and in straight path (sample images in the appendix). The captured intensity (using equation 4) of the ball during motion was converted to position using equation 5. The

position vs. time plot of the acquired positions is shown in Figure 5. A line is fitted on the plot and the equation of the line and the R^2 value is also shown. The slope of the line will then be the velocity of the ball along its path.

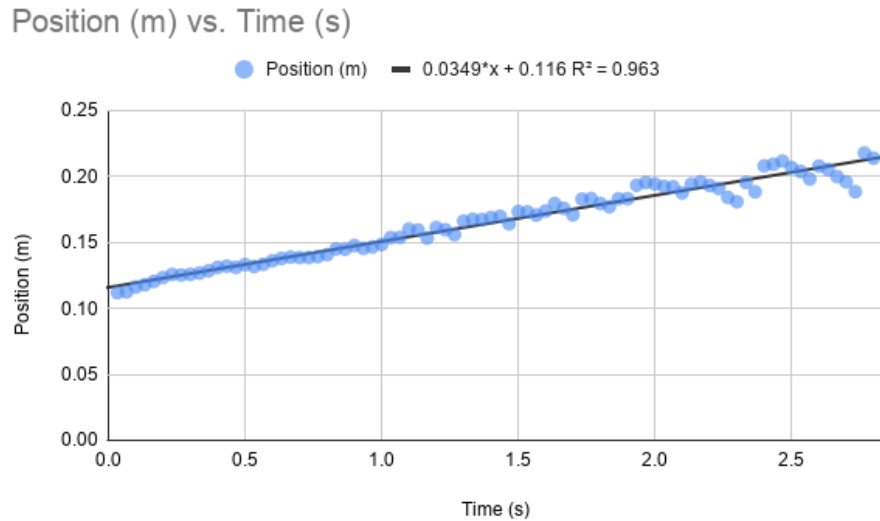


Figure 5: Position vs. Time graph of the ball in fast motion.

Find the derivative of the equation of the trendline. The slope of the line is approximately 0.0349 therefore the velocity of the ball moving in fast motion is 0.0349 m/s. The R^2 value is equal to 0.963 so the results are still acceptable.

For the case of the slow moving ball, 112 frames captures the motion of the ball moving in a straight path (sample images in the appendix). The speed of the ball contributed to the number of data points as the camera captured the motion of the ball for a longer period of time. Again, using equations 4 and 5, the position of the ball is estimated from the intensities in the images. The position vs. time plot for this case is shown in Figure 6.

Position (m) vs. Time (s)

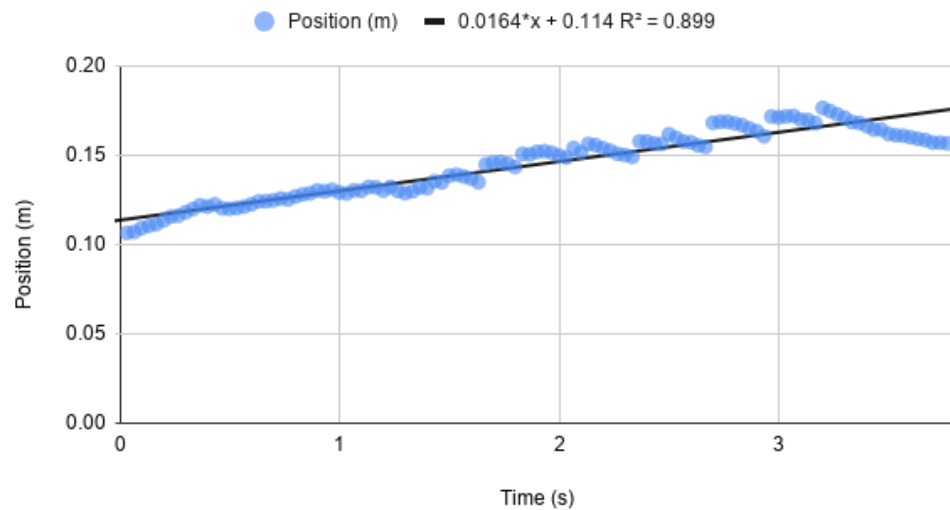


Figure 6: Position vs. Time graph of the ball in relatively slower motion.

Find the derivative of the equation of the trendline. Since the slope of the fitted line is the approximate velocity of the ball, the measured velocity is approximately 0.0164 m/s. It has a relatively lower value compared to the fast moving ball. The fitted line has an R^2 value of 0.899 and is still acceptable.

The recorded velocity of the ball from the stopwatch at slow velocity is 0.0172 m/s and the recorded velocity of the ball at high velocity is 0.0262 m/s. By calculating the percent error of both cases, the percent error for the slow and the fast moving ball are 4.65% and 33.21%, respectively. These errors can percent be accounted by the inaccuracy of the push of the ball or the limitations of the outside observer recording the time. The shorter the time it takes for the ball to reach the standard point, the more inaccurate the time measurement is. This contributes largely to the inaccuracy of the recording thus resulting to a large percent error.

4. Conclusions and Recommendations

Based on the results of the velocity of the fast moving ball and the slow moving ball case, it is concluded that the intensity-based kinematics method is an effective way to calculate the velocity of a moving object. Although there may be restrictions (dark room, solid color object, etc.), this process still shows results with small percent error. The calculated velocity of the slow ball is 0.0164 m/s with a percent error of 4.65% and 0.0349 m/s with a percent error of 33.21% for the fast ball.

Further investigations on the accuracy of this method are recommended. By creating a better setup such that the ball will be moving in constant velocity. It is also recommended to perform several trials to reduce the errors.

References

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- [2] Morphological Transformations. Open CV Image Processing. Retrieved on March 26, 2021 https://opencv-python-tutroals.readthedocs.io/en/latest/py_tutorials/py_imgproc/py_morphological_ops/py_morphological_ops.html#morphological-ops
- [3] Image Segmentation. Wikipedia. Retrieved on March 26, 2021 https://en.wikipedia.org/wiki/Image_segmentation
- [4] Inverse Square Law. HyperPhysics. Retrieved on March 26, 2021 <http://hyperphysics.phy-astr.gsu.edu/hbase/Forces/isq.html>
- [3] Derivative. Wikipedia. Retrieved on April 16, 2021 <https://en.wikipedia.org/wiki/Derivative>

Appendix

Tables and Figures

Table 1: Position and Intensity values for the calibration process

Position	Intensity	Position	Intensity
0.3500	0.1996	0.9500	0.0639
0.4000	0.1854	1.0000	0.0628
0.4500	0.1661	1.0500	0.0608
0.5000	0.1535	1.1000	0.0537
0.5500	0.1324	1.1500	0.0523
0.6000	0.1206	1.2000	0.0412
0.6500	0.1121	1.2500	0.0371
0.7000	0.1058	1.3000	0.0339
0.7500	0.0894	1.3500	0.0314
0.8000	0.0820	1.4000	0.0303
0.8500	0.0695	1.4500	0.0317
0.9000	0.0655	1.5000	0.0278

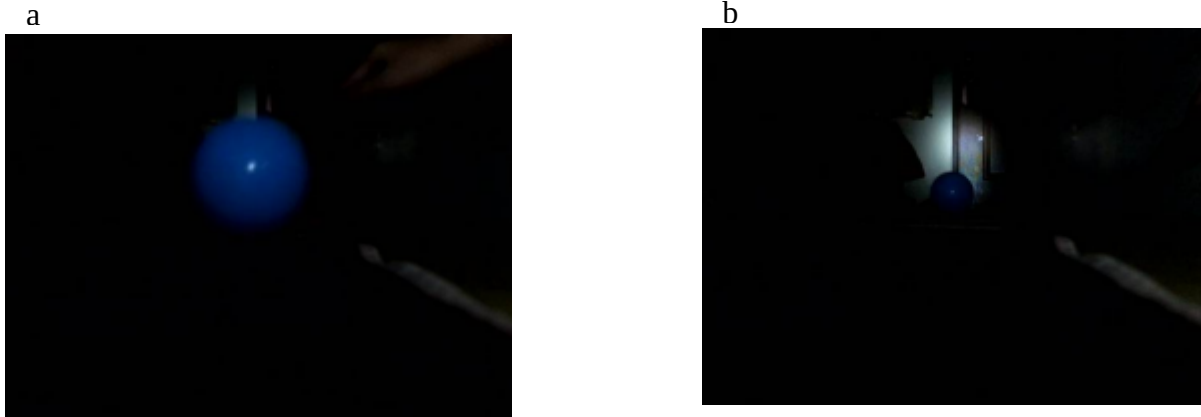


Figure 7: Sample image for the fast moving ball (a) first image (b) image after 83 frames.

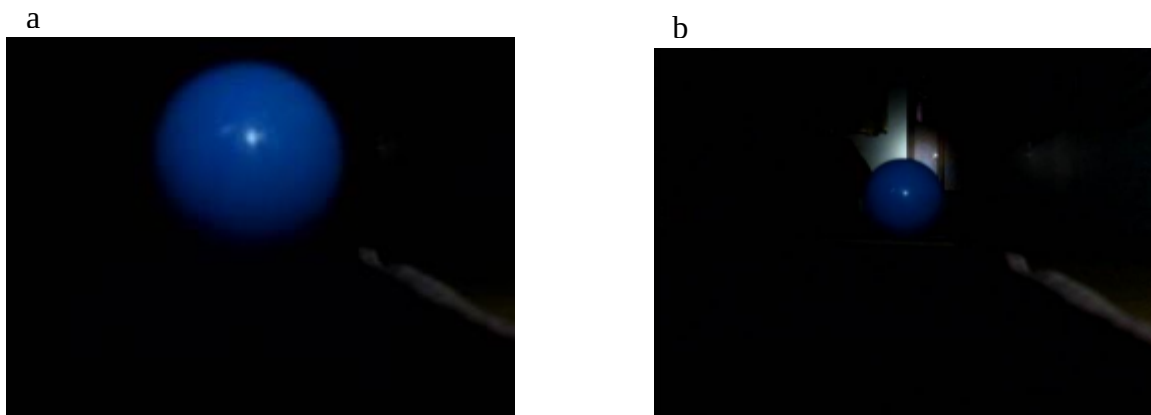


Figure 8: Sample image for the slow moving ball (a) first image (b) after 83 frames.



Figure 9: Sample image of manual image segmentation

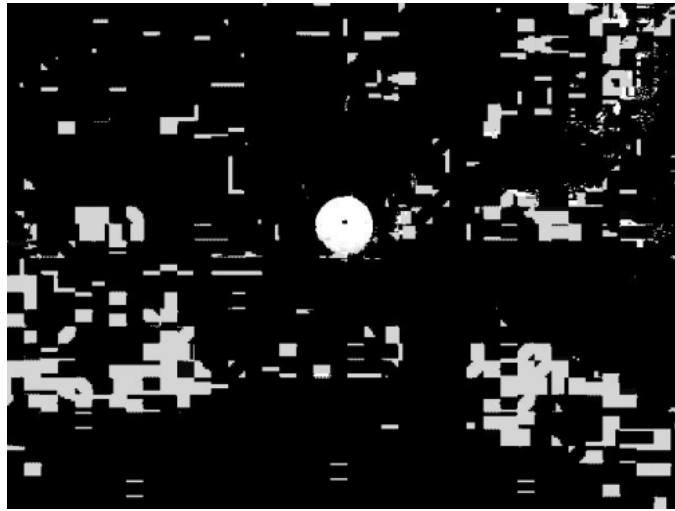


Figure 10: Sample image after performing image segmentation.

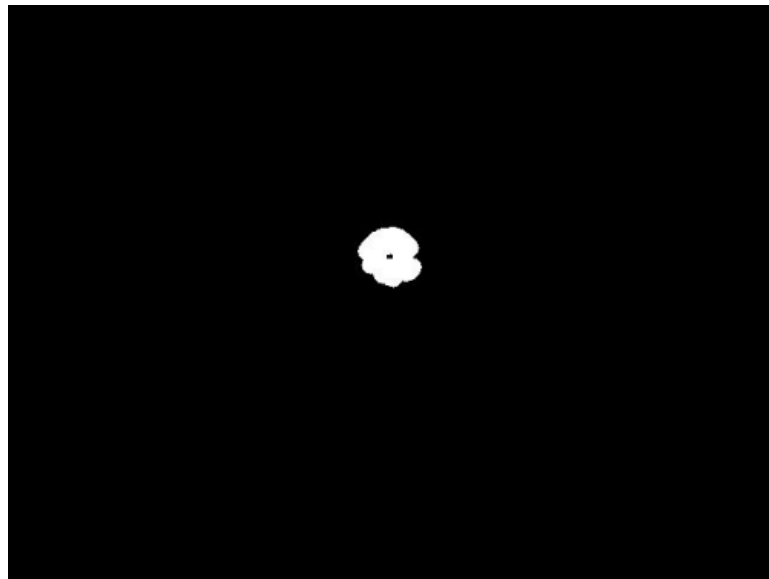


Figure 11: Sample image after performing image segmentation and morphological operations.